Lab 4: Vibe Coding

Chat History and Implementation Record

Repository assigned: https://github.com/SETAPESU26/07_catch_falling

Submission repository: https://github.com/ManishS943/LAB-4---CATCH-FALLING

Date: 1 October 2026

User prompts

Prompt 1:
Attached file: INSTRUCTIONS.pdf
Repository: https://github.com/SETAPESU26/07_catch_falling
Request: Execute this for me using those instructions.

Prompt 2:
Submission repository: https://github.com/ManishS943/LAB-4---CATCH-FALLING

Assistant analysis and responses

The assistant read the supplied PDF and repository README, then identified the required workflow: run the starter game, capture a 10-second before video, complete Tasks 1-4, keep a separate commit for each task, capture a 10-second after video, export the chat history, and upload the Lab-4 folder to the user's personal repository.

The assistant inspected the original source and confirmed the two Task 1 defects: collision detection checked only horizontal position, and the catch loop removed items from the list while iterating. It also confirmed that basket bounds, controlled spawning, and the speed boost were incomplete.

Task-by-task implementation

Task	Prompt interpretation	Implemented result	Commit
1	Fix collision timing and skipped adjacent catches.	Added horizontal and vertical circle/rectangle overlap; replaced in-loop removal with a safe remaining-object pass.	be4fbfe
2	Keep the full basket on screen with smooth held-key movement.	Centralized movement in Basket.move(); clamped the basket center using half its width; retained per-frame held-key input.	40c1204
3	Vary spawn positions and timing, prevent repetition, and cap difficulty.	Added safe spawn lanes, minimum separation, randomized 35-80 frame delays, and a seven-object limit.	2b61739
4	Add a visible temporary speed boost.	Space activates a two-times speed boost for 180 frames; a timer and yellow basket show the active state; speed resets automatically.	b6db551

Verification and deliverables

Automated verification: 11 unit tests passed. The tests cover vertical collision timing, multiple simultaneous catches, both basket boundaries, held movement, spawn safety and separation, the object cap, varied spawn timing, boost activation, and automatic speed reset.

Video evidence: before_changes.mp4 is exactly 10 seconds and visibly reproduces the original premature-score bug. after_changes.mp4 is exactly 10 seconds and visibly demonstrates corrected collision timing, boundary handling, the active boost indicator, and varied controlled spawns.

This PDF records the complete user prompts and the assistant's task-by-task implementation record for the lab. The source code, videos, tests, and this PDF are packaged under Lab-4.